

don't touch. These plates are easier bought than made but making them is so much cheaper. You can find everything you need to build your heated bed at reprap.org/wiki/Heated_Bed

The Electronic Components

Now it's time to discuss the meat of the printer or as the rest of the RepRap community likes to call them, the electronics. Obviously the entire 3D printer would need some sort of firmware to recognise where it needs to print and the objects dimensions. There is a whole list of components available to pick from at reprap.org/wiki/Comparison_of_Electronics

The top three though are the RAMPS (mounted on an Arduino Mega), the Sanguinololu and the Gen7

The RAMPS has been around for a long time and a lot of people have contributed to its development. The Sanguinololu is cheaper and simpler and the Gen7 is a feature rich DIY project on its own. Both the Gen7 and the RAMPS have a whole bunch of add-ons available too.

The stepper motors can be controlled with either Pololu stepper controllers or StepSticks which are mounted on to the boards. There are of course cheaper ones available on Ebay and you can find a list of all the places that sell pre assembled ones at 'reprap.org/wiki/RAMPS#How_to_get_it'

Both boards are plugged in to your computer via a USB cable but if your board comes with a micro SD card then you can print the .stl files directly off the card. If in the event you choose to go for the Sanguinololu, make sure it comes preloaded with an ATmega microprocessor comes with the Sanguinololu boot loader installed. It's possible to load it yourself using a boot loader but it's pretty complicated. The smarter thing to do would be to ask the seller.

The Stepper Motors

The stepper motors are what move the printer across the axes. There are a total of five motors controlled by four controllers. The X & Y axes and the



I like to move it move it.

Extruder all use one motor each and the Z axis is controlled by two. While the rest of the motors may be of different specs, the Z axis motors have to be identical or your Z axis might not move properly.

What the stepper motor does is it runs the motors through tiny steps or micro-steps making it easier to control. The